INCH-POUND
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w/AMENDMENT 3
5 September 2008
SUPERSEDING
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w/AMENDMENT 2
27 March 2008

DETAIL SPECIFICATION

PROJECTILE 155MM, HE, METAL PARTS ASSEMBLY, M107

Inactive for new design after 3 February 1997

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the manufacturing and assembly of the metal parts for Projectile, 155MM, HE, M107 (see 6.1).

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: AMSRD-AAR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490 - Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 - Ammunition Lot Numbering and Ammunition Data Card

MIL-STD-1916 - DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

10535925 - Projectile, 155MM, HE, M107 Metal Parts Assembly

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10535926 - Body Assembly

10535927 - Body

7548993 - Blank, Rotating Band

9280419 - Envelope Drawing, Ultrasonic Inspection Equipment, 155MM, M107

9280420 - Ultrasonic Test Calibration Standard 155MM, M107 Projectile

9280429 - Rotating Band Seat Tester.

(Copies of these drawings may be requested online at Pica.drawing.request@conus.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL (ASTM)

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Apparatus
ASTM E8 - Standard Test Method for Tension Testing of Metallic Materials
ASTM E10 - Standard Test Method for Brinell hardness of Metallic Materials
ASTM E103 - Standard Test Method for Rapid Indentation Hardness Testing of Metallic Materials

(Copies of ASTM standards are available online at http://www.astm.org or from American Society for Testing and materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

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3.1 First article inspection. When specified(see 6.2) a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 10535925.

3.2.1 Body prior to nosing. The materials, components and assemblies shall be in accordance with dwg. 10535927.

3.2.2 Body Machined (prior to band seat machining and threading nose). The materials, components and assemblies shall be in accordance with dwg. 10535927.

3.2.3 Body. The materials, components and assemblies shall be in accordance with dwg. 10535927.

3.2.4 Body Assembly. The materials, components and assemblies shall be in accordance with dwg. 10535926.

3.2.5 Projectile, HE, 155MM,M107 Metal Parts Assy. The materials, components and assemblies shall be in accordance with dwg. 10535925.

3.2.6 Projectile, 155MM, HE, M107 Metal Parts Assembly Destructive Lot Acceptance. The materials, components and assemblies shall be in accordance with dwg. 10535925.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Mechanical properties. Heat treatment or stress relief applied shall be uniform throughout each lot of projectiles and shall be so performed and controlled as to produce in the projectile the mechanical properties specified in 10535927.

3.4.1 Rotating Band. After application and subsequent removal the rotating band shall have the following mechanical properties:

a. Tensile Strength -30,000 lbs per square inch (PSI) minimum

b. Elongation - 25 percent in one inch of gage length, minimum

3.5 Projectile material defects.

3.5.1 Projectile cavity. The projectile cavity shall be free from scale, fins, burrs, draw marks, laminations, imbedded foreign matter and pits. Surface discontinuities which have a fillet or corner radius smaller than .03 inch shall be rejected. (see 6.5).

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3.5.2 Projectile. The Projectile, including the cavity, shall be free from cracks, splits, bursts, cold shuts, pipe, porosity, inclusions, folds, seams, and other metal defects.

3.6 Rotating band.

3.6.1 Foreign matter on band seat. Prior to banding, the band seat shall be free from oil, grease, dirt, rust or other foreign matter.

3.6.2 Application.

3.6.2.1 Destructive. The rotating band shall be applied after final heat treatment of the body, by carefully pressing the band blank into the band seat so as to make contact with the sidewalls for a minimum of eight-tenths of the band seat depth. The band shall be applied in such a matter as not to distort the body wall.

3.6.2.2 Visual. There shall be no gap between the rotating band and the sidewalls of the band seat, after the rotating band has been machined to final dimension.

3.6.3 Band seating. The diametrical clearance between the rotating band and band seat shall not exceed .006 inches, when the destructive method of testing is used or .009 inches when the non-destructive method of testing is used.

3.7 Ultrasonic test. Projectiles shall be free of any discontinuities equal to or in excess of the ultrasonic inspection criteria for the respective zones. The inspection criteria are notches and holes in a standard (on dwg. 9280420) which simulate larger metal defects. Ultrasonic inspection is to be performed in accordance with 9280419 prior to band seat and thread machining.

3.8 Interior cavity finish. The interior cavity after painting shall be free from bare spots, blisters, grease, dirt and other foreign matter. There shall be no pool of paint in the bottom of the cavity.

3.9 Workmanship. The requirements for workmanship are as follows:

3.9.1 Surface preparation. Preparation of metal surfaces shall be as specified on dwg. 10535927. Where cleaning and preparation are required by component details, the finish specified shall apply to the indicated surfaces regardless of contour configuration.

3.9.2 Painting. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

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3.9.3 Burr. No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.9.4 Threads. The threads shall be full and undamaged for the entire minimum length or depth as specified on dwg. 10535927.

3.9.5 Foreign matter. No part of the assembly shall contain dirt, grease, chips, rust, corrosion or other foreign matter.

3.9.6 Protection of rotating band. The machined rotating band shall be protected from being nicked, burred or otherwise damaged in subsequent handling prior to shipment.

3.10 Ballistics test. The projectile assembly shall reveal no evidence of metal parts failure. Metal parts failure shall be defined as:

3.10.1 Breakup. There shall be no loss or breakup of metal parts components in the gun bore or in flight.

3.10.2 Rotating band failure. There shall be no loss or fanning of the rotating band.

3.10.3 Shearing of rotating band. There shall be no shear of rotating band as evidenced by complete lack of engraving.

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4. VERIFICATION

TABLE I. Requirement/Verification Cross Reference Matrix.

Methods of Verification N/A - Not Applicable 1 - Analysis 2 - Demonstration 3 - Examination 4 - Test							Classes of Verification A- Design Verification B- First Article C- Acceptance			
Section 3 Requirements	Description	Verification Method					Verification Class			Section 4 Verification
		N/A	1	2	3	4	A	B	C	
3.1	First Article Test				X	X		X	X	4.2
3.2	Components and assemblies				X	X		X	X	4.3
3.2.1	Body (prior to nosing)				X	X		X	X	4.3.2.1
3.2.2	Body Machined (prior to band seat machining and threading nose)				X	X		X	X	4.3.2.2
3.2.3	Body				X	X		X	X	4.3.2.3
3.2.4	Body assembly				X	X		X	X	4.3.2.4
3.2.5	Projectile, HE, 155MM,M107 Metal Parts Assy.				X	X		X	X	4.3.2.5/4.4.6/ 4.4.5
3.2.6	Projectile, 155MM, HE, M107 Metal Parts Assembly Destructive Lot Acceptance				X	X		X	X	4.3.2.6
3.3	Ammunition lot numbering				X			X	X	4.3.1
3.4	Mechanical properties					X		X	X	4.3.2.3/4.4.2
3.4.1	Rotating band					X		X	X	4.3.2.4/4.4.3.2
3.5.1	Projectile cavity				X			X	X	4.3.2.1/4.3.2.4/4.3.2.5/4.4.1
3.5.2	Projectile				X			X	X	4.4.1
3.6.1	Foreign matter on band seat				X			X	X	4.3.2.4
3.6.2.1	Destructive				X	X		X	X	4.3.2.4/ 4.4.3.3.3
3.6.2.2	Visual				X			X	X	4.3.2.4
3.6.3	Band seating				X			X	X	4.3.2.4/4.4.3.3
3.7	Ultrasonic Test					X		X	X	4.3.2.2/4.4.4

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TABLE I. Requirement/Verification Cross Reference Matrix-Continued

3.8	Interior cavity finish				X			X	X	4.3.2.3/4.3.2.5
3.9	Workmanship				X			X	X	4.3.2.1/4.3.2.3-4.3.2.5
3.10	Ballistic Test					X		X	X	4.3.2.6/4.4.7

4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II. FIRST ARTICLE INSPECTION

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Projectile, 155M, HE, M107 metal parts assembly			
Dwg. 10535925			
Examination for defects	5	3.2.5	4.3.2.5
Ballistic performance for structural integrity	40	3.10	4.4.7/4.3.2.6
Salt spray testing	5	3.2	4.4.6.1
Protective coating thickness	5	3.2	4.4.5
Body assembly			
Dwg. 10535926			
Examination for defects	5	3.2.4	4.3.2.4
Band seating	5	3.6.3	4.4.3.3
Mechanical properties – rotating band	5	3.4.1	4.4.3.2
Body			
Dwg 10535927			
Examination for defects	5	3.2.3	4.3.2.3
Yield strength and elongation test	2	3.4	4.4.2.2

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TABLE II. FIRST ARTICLE INSPECTION - Continued

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Body (prior to band seat machining and threading nose)			
Dwg. 10535927			
Examination for defects	5	3.2.2	4.3.2.2
Ultrasonic test	5	3.7	4.4.4
Body (prior to nosing)			
Dwg. 10535927			
Examination for defects	5	3.2.1	4.3.2.1
Blank, rotating band			
Dwg. 7548993			
Mechanical properties	5	3.2	4.4.3.1

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 (Formation and identification of lots or batches).

4.3.1.1 Heat treatment inspection lot. Heat treatment inspection lots shall not exceed one mill heat of steel. Whenever a break in heat treating of eight (8) hours or more occurs or a furnace is shut down for any length of time, the heat treatment inspection lot will be terminated and a new lot shall start. Additionally, heat treatment inspection lots shall be restricted to components heat treated in the same heat treating equipment, at the same temperature, for the same length of time and in one unchanged process. When a batch type furnace is used, the heat treatment inspection lot shall be further restricted to those components heat treated at the same time in one batch. Batches may be combined into larger heat treatment inspection lots after qualifying with individual batches provided qualifying procedures are submitted to and approved by the Government and provided all other restrictions detailed above are met. Test samples shall be selected from the heat treatment inspection lot after final heat treatment. Components in heat treatment inspection lots shall remain grouped together and identified by inspection lot until mechanical property tests have been completed. Components from heat treatment inspection lots which comply with applicable requirements may be subsequently regrouped into final assembly lots.

4.3.1.2 Miscellaneous Lots. Small lots (such as bodies manufactured from abnormally small heats of steel or bodies that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment lots. Such lots shall be identified by a miscellaneous lot code number and processed in accordance with 4.3.1.1. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.3.1.3 Projectile lots. When it has been determined that bodies of a heat treated lot have complied with the mechanical properties requirement of dwg. 10535927, they may be grouped for further processing.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection

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Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 through 4.3.2.6. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests.

c. Alternative Inspections. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916.

d. Tool control. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.1	**TITLE** Body (prior to nosing)	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 10535927
			NEXT HIGHER ASSEMBLY 10535926

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major 101	Visual indication of metal defects in the body cavity	100%	3.5.1	4.4.1/Visual
Minor 201	Evidence of poor workmanship	Level II	3.9	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.2	**TITLE** Body Machined (prior to band seat machining and threading nose)	**SHEET** 1 **OF** 1	**DRAWING NUMBER** 10535927
			NEXT HIGHER ASSEMBLY 10535926

CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical				
1	Ultrasonic test	100%	3.7	4.4.4
Major				
101	Ultrasonic test stamp missing, see note 1	100%	3.2.2	Visual
Minor	None defined			

1. Any part found without the required test stamp shall be ultrasonically tested to determine if the part fails the critical test. If the part fails the critical test it shall be treated as a Critical Nonconformance. If the part passes the critical test it shall be treated as a Major Nonconformance.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Body	SHEET 1 OF 2	DRAWING NUMBER 10535927
			NEXT HIGHER ASSEMBLY 10535926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Perpendicularity of rear face with body	Level IV	3.2.3	AIE
102	Circular runout of pitch diameter of thread with body and front face	Level IV	3.2.3	AIE
103	Circular runout of ogive with body	Level IV	3.2.3	AIE
104	Circular runout of bourrelet with body	Level IV	3.2.3	AIE
105	Circular runout of boat-tail with body	Level IV	3.2.3	AIE
106	Diameter of rotating band seat	Level IV	3.2.3	AIE
107	Location of rotating band seat	Level IV	3.2.3	AIE
108	Width of rotating band seat	Level IV	3.2.3	AIE
109	Length from front of boat-tail to rear of rotating band seat	Level IV	3.2.3	AIE
110	Depth of knurling	Level IV	3.2.3	Visual
111	Scoring or knurling missing from rotating band seat	Level IV	3.2.3	Visual
112	Diameter of bourrelet, min.	Level IV	3.2.3	AIE
113	Diameter of body	Level IV	3.2.3	AIE
114	Pitch diameter of thread	Level IV	3.2.3	AIE
115	Minor diameter of thread max	Level IV	3.2.3	AIE
116	Thickness through base	Level IV	3.2.3	AIE
117	Thickness of body wall min.	Level IV	3.2.3	AIE

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.3 Cont.	TITLE Body	SHEET 2 OF 2	DRAWING NUMBER 10535927
			NEXT HIGHER ASSEMBLY 10535926

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
118	Variation in wall thickness at ogive and body	Level IV	3.2.3	AIE
119	Total length	Level IV	3.2.3	AIE
120	Length from front face to forward basic diameter	Level IV	3.2.3	AIE
121	Length from front face to rear basic diameter	Level IV	3.2.3	AIE
122	Length of effective thread, min.	Level IV	3.2.3	AIE
123	Length of thread stock , max	Level IV	3.2.3	AIE
124	Mechanical properties test	4.4.2	3.4	4.4.2
Minor				
201	Diameter of thread chamfer	Level II	3.2.3	AIE
202	Angle of thread chamfer	Level II	3.2.3	AIE
203	Radii of chamfers missing or incorrect	Level II	3.2.3	Visual
204	Diameter at rear of boat-tail	Level II	3.2.3	AIE
205	Length from front face to rear of bourrelet	Level II	3.2.3	AIE
206	Surface finish improper	Level II	3.9	Visual
207	Foreign matter in cavity	Level II	3.8	Visual
208	Evidence of poor workmanship	Level II	3.9	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body Assembly	SHEET 1 OF 2	DRAWING NUMBER 10535926
			NEXT HIGHER ASSEMBLY 10535925

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Circular runout of rotating band with body	Level IV	3.2.4	AIE
102	Diameter of rotating band	Level IV	3.2.4	AIE
103	Diameter of rotating band relief groove	Level IV	3.2.4	AIE
104	Location of rotating band	Level IV	3.2.4	AIE
105	Width of rotating band relief groove	Level IV	3.2.4	AIE
106	Any gap between the finish machined rotating band and either sidewall of the band seat	100%	3.6.2.2	Visual/ Dedicated illumination
107	Weight	Level IV	3.2.4	Balance Scale
108	Mechanical properties – rotating band	4.4.3.2	3.4.1	4.4.3.2
109	Rotating band seating	4.4.3.3	3.6.2.1/ 3.6.3	4.4.3.3
110	Visual indication of metal defective	100%	3.5	Visual
111	Foreign matter in band seat prior to banding	100%	3.6.1	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Body Assembly	SHEET 2 OF 2	DRAWING NUMBER 10535926
			NEXT HIGHER ASSEMBLY 10535925

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Minor				
201	Diameter of rotating band groove	Level II	3.2.4	AIE
202	Length from rotating band groove to start of angle	Level II	3.2.4	AIE
203	Length of rotating band flat	Level II	3.2.4	AIE
204	Length total max	Level II	3.2.4	AIE
205	Angle at front of rotating band	Level II	3.2.4	AIE
206	Angle at rear of rotating band relief groove	Level II	3.2.4	AIE
207	Width of rotating band	Level II	3.2.4	AIE
208	Width of rotating band groove	Level II	3.2.4	AIE
209	Depth of undercut front and rear of rotating band (if applicable)	Level II	3.2.4	Visual
210	Width of undercut front and rear of rotating band (if applicable)	Level II	3.2.4	Visual
211	Foreign matter in cavity	Level II	3.2.4	Visual
212	Evidence of poor workmanship	Level II	3.9	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Projectile, HE, 155MM,M107 Metal Parts Assy.	SHEET 1 OF 1	DRAWING NUMBER 10535925
			NEXT HIGHER ASSEMBLY –

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of bourrelet, max (after painting)	100%	3.2.5	AIE
102	Rotating band damaged	Level IV	3.2.5	Visual
103	Foreign matter in cavity	Level IV	3.8	Visual
104	Protective coating thickness	4.4.5	3.2.5	Test
105	Salt spray test	4.4.6	3.2.5	4.4.6.1
106	Pool of paint in cavity	100%	3.2.5	Visual
107	Visual indication of metal defects in body cavity	100%	3.5.1	Visual
Minor				
201	Paint on threads	Level II	3.2.5	Visual
202	Paint on rotating band surface	Level II	3.2.5	Visual
203	Protective coating in cavity inadequate or damaged	Level II	3.2.5	Visual
204	Protective coating on exterior surfaces inadequate or damaged	Level II	3.2.5	Visual
–	Note 1			
206	Marking missing incorrect or illegible	Level II	3.2.5	Visual
207	Grommet missing	Level II	3.2.5	Visual
208	Lifting plug missing	Level II	3.2.5	Visual
209	Evidence of poor workmanship	Level II	3.2.5	Visual

Note 1. This space intentionally left blank

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PARAGRAPH 4.3.2.6	TITLE Projectile, 155MM, HE, M107 Metal Parts Assembly Destructive Lot Acceptance	SHEET 1 OF 1	DRAWING NUMBER 10535925
			NEXT HIGHER ASSEMBLY –

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical				
1	Ballistic test failure, projectile break-up or component separation in-bore or in flight	4.4.7	3.10.1	4.4.7
2	Ballistic test failure, rotating band failure	4.4.7	3.10.2	4.4.7
3	Ballistic test failure, shearing of rotating band	4.4.7	3.10.3	4.4.7
Major	None defined			
Minor	None defined			

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4.4 Testing.

4.4.1 Visual examination for material defects. The internal cavity of the projectile shall be inspected visually prior to nosing using probe lighting to assure conformance with 3.5. Visual standards in accordance with 3.5 shall be set up for use during this inspection. Any projectile failing to meet the applicable requirements of 3.5 shall be classed defective and removed from the lot.

4.4.2 Mechanical properties of the projectile body.

4.4.2.1 Hardness. Each final heat treated projectile from a heat treated lot shall be identified and subjected to a hardness test on the surface between the rear of the rotating band seat to 1/2 inch forward of the rotating band. One (1) hardness reading shall be taken. The hardness reading obtained shall be used as the basis for selecting the tension test sample of paragraph 4.4.2.2.

4.4.2.1.1 Hardness test. The hardness test shall be conducted in accordance with ASTM Method E103 and the following:

a. The surface of the body in the area designated to be hardness tested shall be prepared by grinding a flat to a sufficient depth, .015 inch minimum, to remove scale and get below the decarburization zone. The preparation process to be used shall require approval as part of the AIE submittal. Manual grinding shall not be used. The surface preparation requirement may be eliminated if the conditions of 6.12 are met.

b. Hardness tester periodic verification shall be performed using Brinell test standards manufactured in accordance with ASTM E10 methods in lieu of production samples.

c. The hardness tester shall use a standard 10mm Brinell ball. The body shall be securely supported so that no rocking or shifting of the body will occur during the test. The tester shall apply a preload followed by application of the full load. The preload shall not exceed 10% of the major load of 3000kg. The amount of penetration of the indenter (10mm ball) during that time period when the major load is applied including the dwell time at the end of the load application shall be automatically measured. The tester shall be provided with a means to compensate for any flexing of the test item that may occur during application of the major load. The hardness number will be displayed at the conclusion of the test.

d. One hardness reading shall be taken.

e. All hardness test measurements shall be automatically recorded.

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4.4.2.2 Yield strength and elongation. Two final heat treated projectiles from a heat treated inspection lot shall be selected for the tension test. The projectiles selected shall be those which have the highest and the lowest hardness reading from 4.4.2.1. From each of these projectiles two specimens shall be tested in accordance with 4.4.2.2.1. In the event of rejection, new samples of higher or lower hardness values or both, as applicable, may be selected and tested in order to establish an acceptable hardness range. Once this range has been established all projectiles whose hardness values fall outside the range shall be rejected or subjected to re-temper or heat treat cycle if approved by the Contracting Officer. An acceptable hardness range is one which will provide acceptable yield and elongation results.

4.4.2.2.1 Yield strength and elongation method. The two projectiles selected for testing shall have two specimens taken from each and tested. Two longitudinal specimens shall be taken 180^{0} apart from the sidewall. Location shall be centered under the rotating band seat and in the middle of the wall thickness in that area. Failure of any specimen to comply with the applicable drawing requirement shall be cause for rejection of the heat treated inspection lot. Testing shall be conducted in accordance with ASTM E8 with the largest substandard size round specimen possible.

4.4.2.3 Miscellaneous lots. Miscellaneous heat treatment lots shall be subjected to a hardness test in accordance with 4.4.2.1.1.

4.4.3 Rotating Band.

4.4.3.1 Mechanical properties of rotating band blank. The band shall be cut from front to rear and flattened, without hammering which may further cold work the materiel, and the largest obtainable flat type specimen shall be cut from the band, the length of which shall provide a one inch gage length, plus adequate material at each end to accommodate the grips of the tensile testing machine. This method should yield a .500 inch gage width. In any event, the gage cross section shall be determined to the nearest .001 square inch and utilized to calculate the tensile strength, and elongation. Testing shall be conducted in accordance with ASTM E8.

4.4.3.2 Mechanical properties of disassembled rotating bands. One body assembly shall be randomly selected from each regular production shift or fraction thereof per banding machine in operation and subjected to this test to determine compliance with 3.4.1. Failure to comply shall be cause for rejection of the assemblies represented by the sample. The sample provide shall have the rotating band removed and carefully flattened so as to hold a minimum to any additional cold working. The specimen blank shall be machined so that all serration marks from

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the band seat are removed. A test specimen having a one inch gage length, and grip ends of sufficient length to accommodate the tensile machine shall be removed from the blank and tested in accordance with the procedure prescribed by ASTM Method E-8.

4.4.3.3 Band Seating

4.4.3.3.1 Tightness, non-destructive method. Five body assemblies from each 1-hour production or fraction thereof per banding machine in operation shall be subjected to this test. Failure of one or more bands to comply with 3.6.3 shall be cause for the rejection of the assemblies represented by the samples. The samples provided by 4.4.3.3.1 shall be tested utilizing the equipment and procedures described in drawing No. 9280429.

4.4.3.3.2 Tightness, destructive method. When the non-destructive equipment is not available, one body assembly from each regular production shift or fraction thereof from each banding machine in operation shall be subjected to this test to determine compliance with 3.6.3. Failure to comply shall be cause for rejection of the assemblies represented by the failing sample.

The body assembly selected should not be processed through the machining line, although each band should have a light surface cut to true up the surface prior to measuring. The diameter of each band shall be measured on 3 locations, 60 degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This comprises 6 diametrical measurements per band. These points of measurement shall be marked on the bands and on the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The bands shall be measured at each of the marked points (a total of 12 measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the corresponding 2 thicknesses of the band; the results obtained shall be subtracted from the previously determined diameters over the band.

4.4.3.3.3 Sidewall contact. Rotating bands removed as specified under 4.4.3.2 shall be examined to determine compliance with 3.6.2 Failure to comply shall be cause for rejection of assemblies represented by the sample. The samples shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward and rear of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two other parts, thus exposing four surfaces each of which reveals a view of the band swaged into the band seat. The exposed surfaces shall be subjected to a light grinding operation to remove the metal smear effects of saw

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cuts prior to measurements for depth of sidewall contact to determine compliance with the requirements of 3.6.2.

4.4.3.3.4 Rotating band alternative inspection. A non-destructive test method may be used in place of 4.4.3.2 and the other sub-paragraphs of 4.4.3.3 as described in 6.17.

4.4.4 Ultrasonic inspection. The projectile body shall be ultrasonically inspected prior to band seat machining and nose threading using equipment detailed in 4.4.4.1. Ultrasonic energy shall be used at right angles to assure that orientation does not prevent detection of a defect. Standards, in accordance with 4.4.4.3, shall be used to check the effectiveness of the inspection process. Any projectile failing to meet the requirements of 3.7 shall be classed defective and removed from the lot. All projectiles rejected by ultrasonic inspection shall be visually inspected at the location of the ultrasonic indication. An ultrasonic critical defect shall be defined as any ultrasonic reject that has a visual crack or pipe or a visual defect other than a crack or pipe (lap, fold, seam, etc.) that has an ultrasonic signal that is twice the level of the ultrasonic standard notch for that area. Visual defects other than cracks or piping shall be statically evaluated by a level II (or higher) qualified inspector to determine if a critical defect exists. Visual defects other than cracks or piping with less than twice the level of the ultrasonic standard notch for that area are not considered critical, but shall be rejected as major defects. Critical defect determination will be accomplished within 24 hours of the observation. Stamp the letter "U" on each projectile accepted by ultrasonic inspection.

4.4.4.1 Ultrasonic test equipment and procedures. The ultrasonic inspection shall be performed in accordance with dwg. 9280419 with the use of a liquid to assure a reliable coupling to transmit ultrasonic energy. Should a stream or thin film of liquid be used to couple this energy, controls shall be used to monitor the coupling and assure its effectiveness. Such monitoring is not required if submergence in a liquid is used. Using ultrasonic energy at right angles requires at least two separate sources of ultrasonic energy. This means that in the wall and bourrelet ultrasonic regions, ultrasonic energy shall move within the walls of the projectile with separate significant axial or circumferential components of velocity to detect flaws located on the outside, inside or within the walls. For the base region, significant ultrasonic velocity components shall separately be radial and circumferential. Criteria for detection of cracks are machined notches which simulate actual larger cracks. Details are contained on drawing 9280420. All notches machined on the standard are targets for rejection. That is, any projectile that sends back an echo signal that is greater than or equal to the signal from the standard shall be rejected. Relief portions of radially oriented notches where the full depth approaches the surface are not inspection criteria.

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4.4.4.2 Calibration and operation procedures and equipment. Prior to initiation of ultrasonic inspection, the contractor shall submit, as part of the AIE submittal, for written approval one copy of his proposed procedures, including descriptions of equipment (contractor design) and operation procedures (see 6.3). No change to the procedure shall be made after the original approval without submission and approval of the proposed changes. As a minimum, the adequacy of the inspection process shall be checked by using the standard at the start and finish of each production shift or change in operator, as well as at the shift midpoint during production. All channels shall properly indicate a rejection signal when encountering the notches mentioned in 3.7. If the inspection process is determined to be inadequate, the condition shall be corrected and all items inspected since the last acceptable inspection check shall be re-inspected. Equipment used shall provide for a positive activation of a buzzer or alarm as well as a light system for each channel to alert the operator of the signal indications that are the criteria for rejection. A log book shall be maintained by the operator of each system which is a minimum, documents the use of standards, transducers, instrumentation, repairs, reasons for equipment breakage, if any, and changing of qualified operators or inspectors. This log book shall be made available when requested for Government review.

4.4.4.3 Ultrasonic standards. The ultrasonic inspection standard shall be provided by the contractor for use during the ultrasonic inspection. The standard shall be machined from an acceptable projectile in accordance with drawing 9280420 and shall be inspected for all drawing requirements to determine its acceptability for use. Government verification of all inspections performed on the standard is required

4.4.5 Protective coating - thickness. Two samples shall be selected at random to represent each shift or fraction thereof, of continuous paint production. The sample projectiles shall be inspected for paint coating thickness at a minimum of four separate locations on each projectile. Any projectile failing to meet the applicable requirements shall be classed defective and every projectile painted since the last acceptable sample shall be rejected.

4.4.6 Salt spray. After qualification in accordance with TT-C-490 or commercial equivalent as approved by the contracting officer, two(2) salt spray samples shall be taken each week, one during the first 30 minutes of production and a second at any other time. Failure to comply with the requirements of 3.2.5 shall be cause for taking the corrective action prescribed in TT-C-490.

4.4.6.1 Salt spray testing. Salt spray testing shall be accomplished on production projectiles or sectioned projectiles that were acceptable at the time of painting and were sectioned subsequent to painting. If sectioned projectiles are used,

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sectioning shall be accomplished to allow testing of all major areas of the projectiles on an alternating basis. The salt spray test shall be performed utilizing the equipment and procedures specified by ASTM B117 and TT-C-490 or commercial equivalent as approved by the procuring contracting officer, with the exception that test samples shall be selected in accordance with 4.4.6.

4.4.7 Ballistic performance test for structural integrity. These tests shall be performed at a Government proving ground or Government approved installation in accordance with the applicable acceptance test procedure. The sample projectiles shall be inert loaded, assembled and fired at ambient temperature in an M185 or M199 series cannon with 25% minimum tube life remaining using an M203 series propelling charge. The propelling charges shall be uniformly temperature conditioned and adjusted to provide a chamber pressure of 41,000 + 3000 PSI. Camera and radar coverage shall be provided to observe for projectile failure from the gun muzzle to the target area. Note that camera coverage need only be obtained for the muzzle region to record the condition of the projectile as it exits the tube. Muzzle velocity and chamber pressure shall be recorded for information purposes. Visual observation shall be made to assure that the projectile impacts in the target area. Range and deflection shall be recorded for informational purposes only. In the event of any abnormal condition, as observed by personnel or instrumentation, an attempt shall be made to recover and examine the metal parts.

4.4.7.1 Initial inspection. Beginning with the first lot produced and continuing until three consecutive lots have complied with requirements specified, 30 sample projectiles shall be randomly selected for this test. The 30 sample projectiles shall be observed for structural integrity of parts when tested in accordance with 4.4.7. If any sample projectiles fail to comply with the requirements for structural integrity of parts, the lot shall be rejected.

4.4.7.2 Reduced inspection. After three consecutive lots have complied with 4.4.7.1, 10 sample projectiles shall be randomly selected from each lot for this test. The 10 sample projectiles shall be observed for structural integrity of parts. If any of the sample projectiles fail to comply with the requirement for structural integrity of parts, the lot shall be rejected. If the lot is rejected the inspection criteria reverts back to paragraph 4.4.7.1.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of

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Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The projectile covered by this specification is intended for use on the Projectile, 155MM, HE, M107. This specification contains requirements suitable for production.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. For acquisition purposes, the contract or order should specify packaging requirements. The projectiles should be palletized and marked in accordance with drawing 12944446 or alternative 9362569 and be ordered by the applicable stock number.

c. Requirements for submission of first article sample.

d. Submission of contractor acceptance inspection equipment (AIE) (see 6.3).

e. Provisions for Certification of nondestructive test (NDT) personnel (see 6.4).

f. Establishment of visual defect standards (see 6.5).

g. Submission of ballistic test data and ammunition data cards (see 6.6).

h. Provisions for critical characteristic controls (see 6.9).

i. Provisions for projectile cavity characteristic controls (see 3.6.1).

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.3.1 Inspection of band seat.

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6.3.1.1 Rotating band seat location and runout. The Contractor may use any control method that will assure the band seat meets the dimensional requirements of the applicable drawing. The inspection plan must include all the required criteria to assure the dimensional aspects of the drawing will be equal to the characteristics defined in 4.3.2.3. When the contractor establishes a control method, the manufacturing procedure, and inspection plan should be forwarded to the Technical Agency as part of the AIE submittal prior to use in accordance with 6.3. No change to these controls will be made once established without approval from the contracting officer.

6.4 Personnel qualification and certification for non-destructive testing (NDT). Personnel performing the ultrasonic testing should be qualified and certified in accordance with recommended practice NAS-410 and procedures approved by ARDEC. Minimum qualification levels should be as follows:

Level III: Review and approval of NDT procedures, equipment and personnel certification plus the following;

Level II: Set-up, calibration, modification of NDT equipment and procedures plus the following;

Level I: Operation of NDT equipment.

6.5 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first month of production. However, establishment of visual standards will not waive the requirements for compliance with 3.5.

6.6 Submission of ballistic test data and ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ballistic test data and ammunition data cards should be forwarded to Commander, TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, and AMSTA-AR-FSA-M, Picatinny Arsenal, NJ 07806-5000, and AMSMC-QAD-I Army Master Depository, Rock Island, IL 61299-6000.

6.6.1 Range and deflection. Range and deflection are not technical requirements of this specification. They are mandatory performance requirements that are achieved by design and are affected by variability in the manufacturing process that may not be completely controlled by the present TDP.

6.7 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot

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formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

6.8 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

6.9 Critical characteristic controls. All production contracts should contain requirements for handling critical defects. As part of all proposals, suppliers should be asked to describe policies, procedures and controls for all operations associated with safety characteristics, how they are documented and maintained under the supplier's integrated management system.

6.10 Steel quality.

a. It is the responsibility of the manufacturer of the projectile metal parts to select a fully-killed carbon steel of a grade compatible with the abilities of the equipment and the methods of processing the metal within the operation of the shops of the forger and machiner to produce the projectile to the dimension and mechanical properties and metal integrity specified as an end item requirement.

b. The supplier of the selected steel raw materiel is responsible only for furnishing a metal composition, grade and, type with dimensions, weight, soundness and freedom from metallurgical or mechanical defect limitations identifiable with the standard AISI grade selected by the projectile manufacturer and complying with the specific ASTM designation and tests selected by the projectile manufacturer.

c. In the application of carbon steel, blooms, billets, slabs and bars, forging quality processing techniques adapted to use of the selected steel should be applied with the expectation that there will be variables in the metal cross-sectional dimension in weight per unit length, in the hardenability variance from heat to heat, in the mechanical condition of the surface, and in the straightness of the raw material. Economic advantages will be realized if the process and equipment within the shops of the forger and the machiner have been engineered in terms of equipment sequence of operations, tooling designs, handling devices and processing techniques having capabilities of compensating for such variables.

6.11 Definitions

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international

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standard which is used to assure conformance of material to the contract requirements.

b. heat treatment. Heat treatment is understood to be any thermal process which brings about a modification in the mechanical and metallurgical properties of the metal involved.

6.12 Hardness test. The surface preparation requirement of 4.4.2.1.1 may be eliminated if the following conditions are met:

(1) Lab results confirm that current production process produces a decarburized zone of .003 inch maximum in the test area. The contractor should provide samples, or photo-micrographs, of high and low hardness samples to the Government for approval. Samples may be the same as for 4.4.2.2.

(2) A controlled atmosphere is maintained in the austenitizing furnace.

6.13 Critical classification notes.

6.13.1 Ultrasonic test. Metal parts defects, except rotating band gaps, large enough to fail by ultrasonic test of the explosive loaded projectile are likely to cause premature functioning from break-up of the projectile in the gun during firing.

6.13.2 Ballistic lot acceptance test. Any premature projectile function or metal parts separation in the gun or in flight is likely to cause injury or death for the Soldier firing or supported by the howitzer.

6.13.3 Rotating band failure. Loss of a rotating band in flight or in bore may result in projectile instability and subsequently a short round potentially resulting in injury or death for Soldiers in the field.

6.13.4 Shearing of rotating band. A sheared rotating band may result in a failure of the projectile to fully spin up, resulting in projectile instability and subsequently a short round potentially resulting in injury or death for Soldiers in the field.

6.14 Packaging Drawings. Drawings listed in this section are for reference only. The government's current packaging for the M107 Metal Parts Assembly as defined by Technical Data Package (TDP) 9362569 and 12914619, is included for reference. If the contractor chooses to build to the current approved TDP (9362569 and 12944449) the Government will waive the testing required to meet the First Article performance test requirements of the packaging specification. The contractor may use the

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information contained in the TDP as guidance for its own design. Unless otherwise noted, the custodian of the following drawings is the U.S. Army Tank-Automated Command- Armament Research, Development and Engineering Center, Picatinny Arsenal, NJ 07806-5000;

9362569	- Packing & Marking for Pallet, Wood Ammo for PJCTL 155MM
12944449	- Packing and Marking for 155MM Projectile pallet

6.15 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

6.16 Subject term (keyword) listing.

Reduced inspection
Salt spray
Ultrasonic test

6.17 Rotating band alternative inspection. The contractor and the government may agree upon band swaging process controls which have been shown to produce acceptable rotating bands. The manufacturing procedure, the inspection plan for controlling the process and any requirements on the rotating band should be part of the AIE submittal prior to implementing into production. Once agreed upon these controls should not be changed without prior Government approval. The contractor may submit a non-destructive test procedure including calibration standard and non-destructive test equipment (see 6.17.3). The inspection of the rotating band should cover the aft end of the band if the entire band cannot be inspected. This alternative method may be implemented, with government approval once the following conditions have been met: Two body assemblies (one sample) as provided under 4.4.3 should be randomly selected from each 4-hour production or fraction thereof from each banding machine in operation to determine compliance with the requirements of 3.4.1, 3.6.2.1 and 3.6.3. Failure to comply may be cause for rejection of the assemblies represented by the sample. A banding machine is considered qualified after 40 consecutive samples have been found acceptable.

6.17.1 Alternative inspection sampling. One body assembly from each 1-hour production or fraction thereof per banding machine in operation should be subjected to this test. While the alternative test procedure is in place the original inspection

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criteria of 4.4.3 should not be followed. Failure of one or more bands to comply with the agreed upon criteria may be cause for the rejection of the assemblies represented by the samples.

6.17.1.1 Alternative inspection testing. The samples provided by 6.17.1 should be tested for conformance to 3.6.3 following the outline of 6.17. As an alternate, the samples may instead be tested following drawing No. 9280429.

6.17.1.2 Alternate inspection test failure. In the event of a non destructive testing failure or a failure to meet the visual gap requirement of 3.6.2.2, rotating band swaging should immediately cease on the banding machine which produced the failure. The part in question should be destructively tested per 4.4.3.3.2 to verify the gap. All bands since the last inspection should be non-destructively tested for acceptance. An investigation into the cause of the failure should be launched. Once determined and addressed swaging should resume with the first 10 bands being inspected. If all ten samples are acceptable, inspection may return to post qualification sampling.

6.17.2 Process changes. If the process changes for any reason the new process should be re-qualified before the sampling of paragraph 6.17.1 may continue.

6.17.3 Submission of rotating band swaging procedures. The procedures for applying the rotating band and for maintaining, controlling and monitoring the rotating band application should be submitted to: Commander, ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, New Jersey 07806-5000 for review and concurrence.

6.17.4 Rotating band mechanical properties. In the case of the non-destructive test, the requirements of paragraph 3.4.1 should not be tested for. However these requirements should instead be applied to the incoming band blanks and tested prior to swaging.

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